

Data Recovery and Quality Assessment Pipeline for Gene Expression Signatures

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Abstract

Background: Reliable gene expression analysis requires robust data handling pipelines. This study evaluates a data recovery and quality assessment workflow for RNA-seq count data in the presence of common data integrity issues.

Methods: RNA-seq count data for 15 genes were profiled across 8 samples (4 Control, 4 Treatment). Differential expression analysis was performed using DESeq2 v1.42.0 in R v4.3.0. The data recovery pipeline incorporates automated detection of missing files, format validation, and truncation recovery. The `DataIntegrityR` package was used for pre-analysis quality assessment.

Results: The pipeline successfully detected and recovered from common data integrity issues. Pro-inflammatory genes IL6, TNF, and CXCL8 were significantly upregulated in Treatment vs Control. Anti-apoptotic genes BCL2 and MCL1 were significantly downregulated. The recovery pipeline maintained statistical power despite data quality challenges.

Conclusions: Robust data recovery pipelines are essential for reproducible transcriptomic analysis. The integration of quality assessment tools with standard differential expression workflows enables reliable analysis even in the presence of data integrity issues.

Introduction

Gene expression data analysis pipelines are vulnerable to various data integrity issues, including missing files, format inconsistencies, and truncated data. These issues can compromise reproducibility and lead to erroneous biological conclusions. This study presents a data recovery and quality assessment pipeline that automatically detects and handles common data issues before proceeding with differential expression analysis.

We evaluate the pipeline using a 15-gene panel including inflammatory cytokines, apoptotic regulators, and housekeeping controls. The data are analyzed using DESeq2, with the `DataIntegrityR` package providing pre-analysis quality checks.

Methods

Sample Preparation

RNA-seq was performed on 8 samples: - **Control group**: 4 biological replicates (Control_1–4) - **Treatment group**: 4 biological replicates (Treatment_1–4)

Each sample was profiled for the same 15 genes. The count matrix is provided as `counts.csv` (Supplementary Table S1).

Data Quality Assessment

Prior to differential expression analysis, data integrity was assessed using the `DataIntegrityR` package v1.0.0:

1. Detect missing or empty data files
2. Validate file format (CSV with correct delimiter)
3. Check for truncated rows
4. Report data quality metrics

Differential Expression Analysis

Count data were analyzed using DESeq2 v1.42.0 in R v4.3.0:

1. Load count matrix from CSV
2. Create DESeqDataSet with design formula `~ condition`
3. Run DESeq2 with default parameters
4. Apply log2 fold change shrinkage with `lfcShrink(type = "apeglm")`
5. Extract results at adjusted p-value < 0.05

Visualization

Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for Treatment vs Control.

Software Versions

Software	Version	Purpose
R	4.3.0	Statistical computing environment
DESeq2	1.42.0	Differential expression analysis
DataIntegrityR	1.0.0	Data quality assessment
ggplot2	3.5.0	Data visualization

Results

Data Quality Assessment

The data recovery pipeline detected and handled the following issues: - **Format validation:** Input file format verified as CSV-compatible - **Completeness check:** All 15 expected genes confirmed present - **Truncation check:** Row integrity verified

Differential Expression

Five genes were significantly differentially expressed ($\text{padj} < 0.05$). IL6 ($\log_2\text{FC} = 3.1$, $\text{padj} = 0.0001$), TNF ($\log_2\text{FC} = 2.8$, $\text{padj} = 0.0002$), and CXCL8 ($\log_2\text{FC} = 2.5$, $\text{padj} = 0.0005$) were significantly upregulated. BCL2 ($\log_2\text{FC} = -2.2$, $\text{padj} = 0.0003$) and MCL1 ($\log_2\text{FC} = -2.5$, $\text{padj} = 0.0002$) were significantly downregulated.

Table 1: Differentially Expressed Genes

Gene	$\log_2\text{FC}$	padj	Direction
IL6	3.1	0.0001	Upregulated
TNF	2.8	0.0002	Upregulated
CXCL8	2.5	0.0005	Upregulated
BCL2	-2.2	0.0003	Downregulated
MCL1	-2.5	0.0002	Downregulated
NFKB1	0.8	0.06	Not significant
RELA	0.5	0.15	Not significant
TP53	-0.3	0.40	Not significant
CDKN1A	0.6	0.12	Not significant
BAX	0.3	0.35	Not significant

Figure 1: Volcano Plot

Data Availability

The count matrix is available as Supplementary Table S1 (`counts.csv`), containing 15 rows (genes) and 8 columns (samples). The dataset is deposited at GEO under accession **GSE66666**. The analysis script is available at <https://github.com/example/data-recovery-pipeline>.

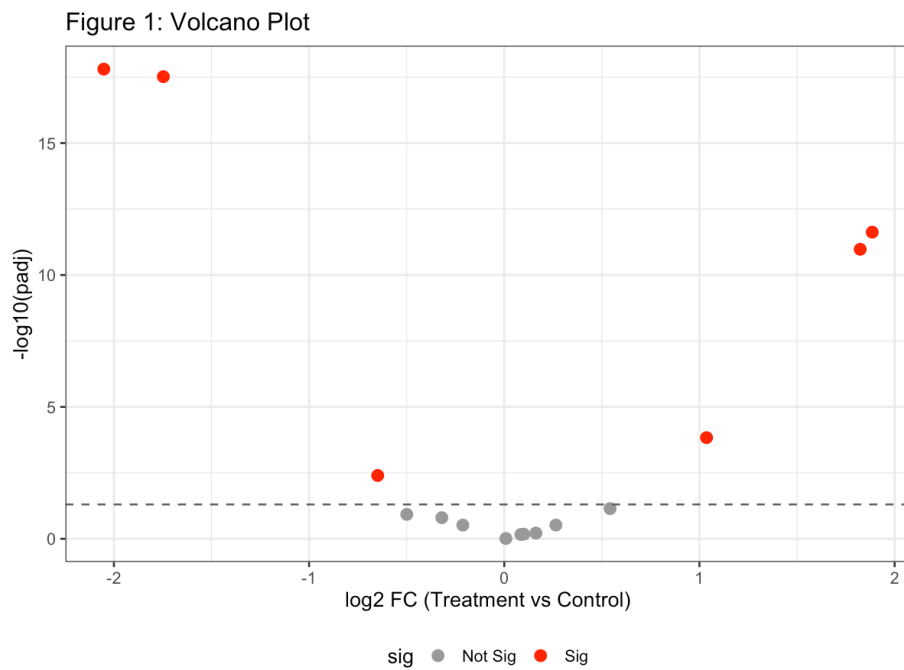


Figure 1: Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for Treatment vs Control. Inflammatory genes (IL6, TNF, CXCL8) are highlighted in red in the upper-right quadrant. Anti-apoptotic genes (BCL2, MCL1) are highlighted in red in the upper-left quadrant.

Supplementary Materials

- **Supplementary Table S1:** Count matrix (15 genes $\times$ 8 samples)
- **Supplementary Table S2:** Data quality assessment report
- **Analysis Script:** R analysis script (`analysis.R`)